



In situ formed three-dimensional (3D) lithium–boron (Li–B) alloy as a potential anode for next-generation lithium batteries

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Abstract The practical application of the lithium anode in lithium metal batteries (LMBs) has been hindered by the uncontrollable growth of lithium dendrite and the high volumetric change during cycling. Herein, the in situ formed three-dimensional (3D) lithium–boron (Li–B) alloy is suggested as an excellent alternative to the Li metal, in which the 3D LiB skeleton can mitigate the growth of Li dendrites and volumetric change. In this study, the Li–B alloy anodes with different B contents were manufactured by high-temperature melting. It was found that the boron content had a significant effect on the electrochemical performance of the Li–B alloy. The Li–B alloy with the least B content (10 wt%, 10LiB) demonstrated the lowest overpotential of 0.0852 V after 300 h and the lowest interface resistance. However, the full cell with 15LiB as the anode displayed the best cycling performance of 115 mAh·g^{−1} after 100 cycles with a coulombic efficiency greater than 97%. The obtained results suggest that the in situ formed three-dimensional Li–B alloy anode can be an excellent alternative to the Li anode via tuning B contents for next-generation high energy density LMBs.

Keywords Li–B alloy; 3D skeleton; Solid electrolyte interface (SEI); Anode; Lithium metal battery

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1 Introduction

With the soaring demand for energy storage in the advanced electronic devices and energy vehicles, it becomes increasingly crucial to develop high-power and high-performance next-generation energy storage systems [1, 2]. The widely used lithium-ion batteries (LIBs), which were commercialized by Sony Corporation in 1991, have almost attained their energy density bottleneck after nearly three decades of development [3–6]. This, in partial, can be attributed to the graphite anode which has a limited specific capacity of 372 mAh·g^{−1}. Researchers are now eagerly looking for next-generation electrode materials which can outperform the currently used graphite electrode materials. Li metal is regarded as the “Holy Grail” electrode material, due to its extremely high theoretical specific capacity (3861 mAh·g^{−1}) and low negative potential (−3.04 V (vs. standard hydrogen electrode)). However, the practical application of the Li metal anode is seriously impeded by the formation of Li dendrites, which could penetrate the separator to cause safety problems such as short circuit, fire and explosion [7]. To mitigate the growth of Li dendrites, three categories of strategies have been proposed: (1) optimizing the electrolyte using different additives [8–10]; (2) building flexible artificial solid electrolyte interface (SEI) film to prevent dendrite formation [11, 12]; and (3) using three-dimensional (3D) structured current collectors or nanostructured Li to lower the actual current density to decrease the likelihood of Li dendrite growth [13–17]. Among these methods, designing different types of 3D current collectors seems to be an efficient and simple way to suppress the generation of Li dendrites. For example, it has been reported that customized 3D nanostructured Cu current collector [18], carbon-coated copper foil [19] and



self-supporting carbon skeleton [20] can effectively suppress the formation of Li dendrites. However, the manufacturing of these nanostructured 3D scaffolds is a complex and expensive process. In addition, these 3D current collectors do not contain Li, and a large amount of Li is required during battery assembly. Consequently, the integrated 3D electrode containing sufficient amount of Li is highly desirable for the Li metal batteries.

The Li–B alloy is suggested as a promising alternative to the Li metal anode because of the following aspects. Firstly, the Li–B alloy has physical properties similar to Li, and the density of the LiB compound is only $1.5 \text{ g}\cdot\text{cm}^{-3}$, which is much less than that of Cu ($8.9 \text{ g}\cdot\text{cm}^{-3}$) [13, 15]. In addition, the manufacturing technology of the Li–B alloy is relatively mature and economically feasible [21–23]. Secondly, the Li–B alloy possesses a negative potential similar to Li metal ($\sim 0.4 \text{ V}$ higher than Li/Li^+), which is favorable for assembling high-voltage batteries. Previous reports have also demonstrated excellent electrochemical performance of the Li–B alloy in Li metal batteries [24, 25]. The Li–B alloy can effectively prevent the growth of Li dendrites, due to the reduction in the actual current density and interface impedance, thereby improving the cycle performance significantly [26]. The LiB fibers form a 3D network skeleton in the Li–B alloy which has good electrical conductivity and lithiophilic property. The LiB skeleton with a certain rigidity can inhibit the dramatic volumetric change during cycling [25, 27]. Therefore, Li–B alloys are regarded as one of the most promising anodes to replace pure Li [26, 28]. Moreover, the low polarization and outstanding cycle stability of Li–B alloy batteries have been demonstrated [13, 15, 29]. Nevertheless, there is an irreversible capacity decay after the long-term electrochemical cycling of the Li–B alloy assembled batteries; the underlying cause remains elusive. Consequently, further studies aimed at elucidating the degradation mechanism are highly desirable.

Herein, the in situ formed three-dimensional Li–B alloy was prepared and its electrochemical performance as an anode material was studied. Specifically, the compositional, morphological and electrochemical properties of the prepared Li–B alloy anode were thoroughly investigated. It was discovered that a dead SEI deposition layer would be generated on the surface of the alloy anode after electrochemical cycling, which is intimately correlated with the performance decay of batteries. Furthermore, it was found that the amount of the generated dead SEI depositions can be decreased by tuning the content of B in the alloy. An asymmetric cell with the B content of 10 wt% (marked as 10LiB) displays the lowest overpotential and interface resistance. However, the full cell with 15LiB (15 wt% B) as the anode demonstrates better cycling performance after 100 cycles with a coulombic efficiency greater than 97%.

This study indicates that the in situ formed three-dimensional Li–B alloy anode is a potential alternative to the Li anode in next-generation high energy density LMBs.

2 Experimental

The detailed procedure of manufacturing the Li–B alloy was previously reported by the group of Liu [21]. Specifically, the alloy was prepared by melting the pure Li metal (purity > 99.9 wt%) and amorphous boron powder (purity > 96.0 wt%) in an iron crucible under an Ar atmosphere at a predetermined ratio. Subsequently, the ingot was rolled to obtain flat strips with a thickness of 0.2 mm. The strips were punched into disks with a diameter of 16 mm. Three samples were named according to the B content as follows: 26LiB, 15LiB and 10LiB for the Li–B alloys with 26 wt%, 15 wt% and 10 wt% B, respectively.

Electrochemical performances were evaluated using the CR2016 coin-type cells. The carbonate electrolyte which contains $1.0 \text{ mol}\cdot\text{L}^{-1}$ lithium hexafluorophosphate (LiPF_6) dissolved in ethylene carbonate (EC), dimethyl carbonate (DMC) and ethyl methyl carbonate (EMC) at a volumetric ratio of 1:1:1 was used. The cells used Celgard 2400 as the separator. The coin cells were assembled in an Ar-filled glove box with the water and oxygen contents below 0.1×10^{-6} . The coin-type asymmetric cell was assembled with the Li–B alloy as the work electrode and Li as the counter and reference electrodes. The coin-type full cell was assembled with the delithiated $\text{Li}_{0.5}\text{CoO}_2$ cathode and Li–B alloy anode. The mass ratios between the anode and cathode of the 26LiB, 15LiB and 10LiB cells were 1.38:1, 1.16:1 and 1.04:1, respectively. The constant current charging/discharging and cycle tests were conducted on the Neware battery test system at room temperature.

To avoid oxidation, the preparation of the X-ray diffraction (XRD) sample was completed in a glove box filled with Ar, and the sample was sealed with a plastic film. The XRD pattern of the Li–B alloy was obtained by X-ray diffractometer (D/Max 2500 VB, Rigaku Corporation, Japan) with $\text{Cu K}\alpha$ radiation. The scanning speed was $8^\circ\cdot\text{min}^{-1}$, and 2θ range was from 20° to 90° . The surface morphology of the electrodes was obtained by field emission scanning electron microscope (FESEM, Quanta 650 FEG, FEI, Czech Republic). The cells were disassembled in the Ar-filled glove box. The harvested Li–B alloy electrode samples were then rinsed with the DMC solvent and sealed in plastic bags after drying. Electrochemical impedance spectroscopy (EIS) was performed using a Princeton electrochemical workstation in the frequency range of 1×10^5 to $1 \times 10^{-2} \text{ Hz}$ with an amplitude of 5 mV. The change of the SEI generated on the Li–B alloy surface was studied in the Li–B/Li cells. The first charge

capacity was 27 mAh, which was approximately half of the total free Li capacity. Subsequently, the charging/discharging cycle test was performed with a limited capacity of 3 mAh. The current density was $0.5 \text{ mA}\cdot\text{cm}^{-2}$.

3 Results and discussion

3.1 XRD pattern of Li–B alloy

Figure 1 illustrates the XRD pattern of the obtained 26LiB alloy. It can be seen that the Li–B alloy comprised hexagonal LiB phase (JCPDS No. 52–1033) and cubic Li phase (JCPDS No. 15–0401). The space group of the LiB compound is No.194, $P_{63/m}^2 2_1 2_1$, and the lattice constants are $a = 0.4022 \text{ nm}$ and $c = 0.2796 \text{ nm}$ [22]. Some diffraction peaks were absent, such as peaks at 41.3° , 62.7° and 66.8° , due to the rolling process, and the weakening of the diffraction peak occurred at the peak with crystal face index (hkl) and $l \neq 0$ [22].

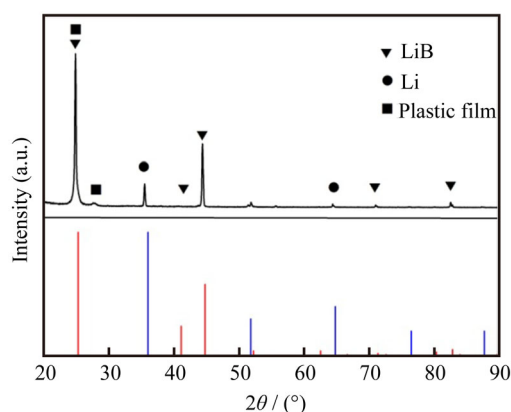


Fig. 1 XRD pattern of obtained Li–B alloy (26LiB)

3.2 Li dissolution/deposition analysis of Li–B alloy

The Li dissolution mechanism of the obtained Li–B alloy was studied by electrochemically dissolving the Li from the Li–B alloy in the Li–B/Li cells at a current density of $0.5 \text{ mA}\cdot\text{cm}^{-2}$. Simultaneously, ex-situ XRD measurement of the Li–B electrode was conducted at specific stages during the dissolution process. Figure 2a illustrates the evolution of the XRD pattern of the Li–B alloy during the Li dissolution process. Initially, it can be seen from Fig. 2a that the main peak intensity of $2\theta = 36.190^\circ$ belonging to Li continuously decreased, which implies that free Li was gradually stripped from the Li–B alloy anode during the dissolution process. When the capacity of the Li dissolution process reached $\sim 50 \text{ mAh}$ (at the voltage platform of $\sim 0.40 \text{ V}$), the main peak of Li was completely disappeared, indicating that free Li had been completely dissolved from the Li–B alloy anode. The calculated discharge capacity of free Li is $\sim 1187.6 \text{ mAh}\cdot\text{g}^{-1}$. Subsequently, the main peak intensity of $2\theta = 45.062^\circ$ corresponding to the LiB compound started to decrease in the XRD pattern and three discharge plateaus of the LiB compound appeared in the electrochemical curve. Further electrochemical dissolution of Li caused the extraction of Li from the LiB compound, which primarily led to the change of the lattice constant c in the LiB compound due to the reduction in the Li atoms in the LiB compound crystal structure [23]. From the above-mentioned results, it can be stated that the free Li filling the pores of the LiB compound skeleton was initially to be dissolved during the Li dissolution process, followed by the dealloying of the LiB compound skeleton. Both free Li and the alloyed Li can be extracted from Li–B alloy. Figure 2b illustrates the evolution of the XRD pattern of the Li–B alloy during the Li deposition process for 50 h. With the deposition of Li, the peaks of Li and the LiB compound gradually reappeared, demonstrating that the electrochemical Li dissolution/deposition in the LiB compound skeleton was reversible. This implies that a part of

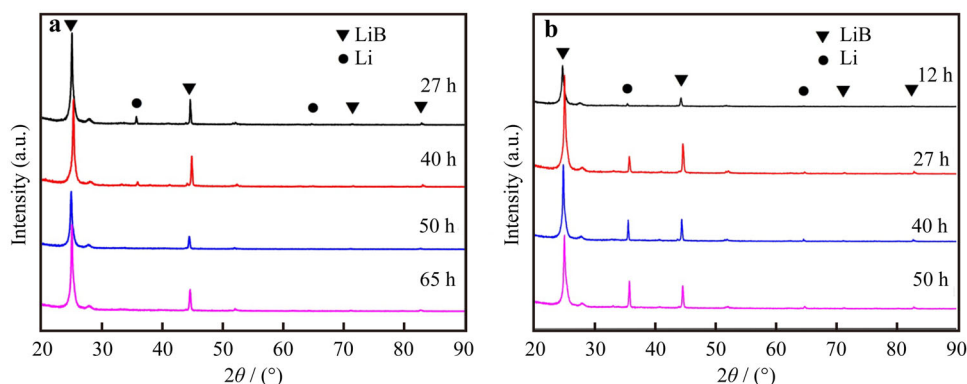


Fig. 2 XRD patterns of Li–B alloy anode during a Li dissolution and b Li deposition at different time at current density of $0.5 \text{ mA}\cdot\text{cm}^{-2}$

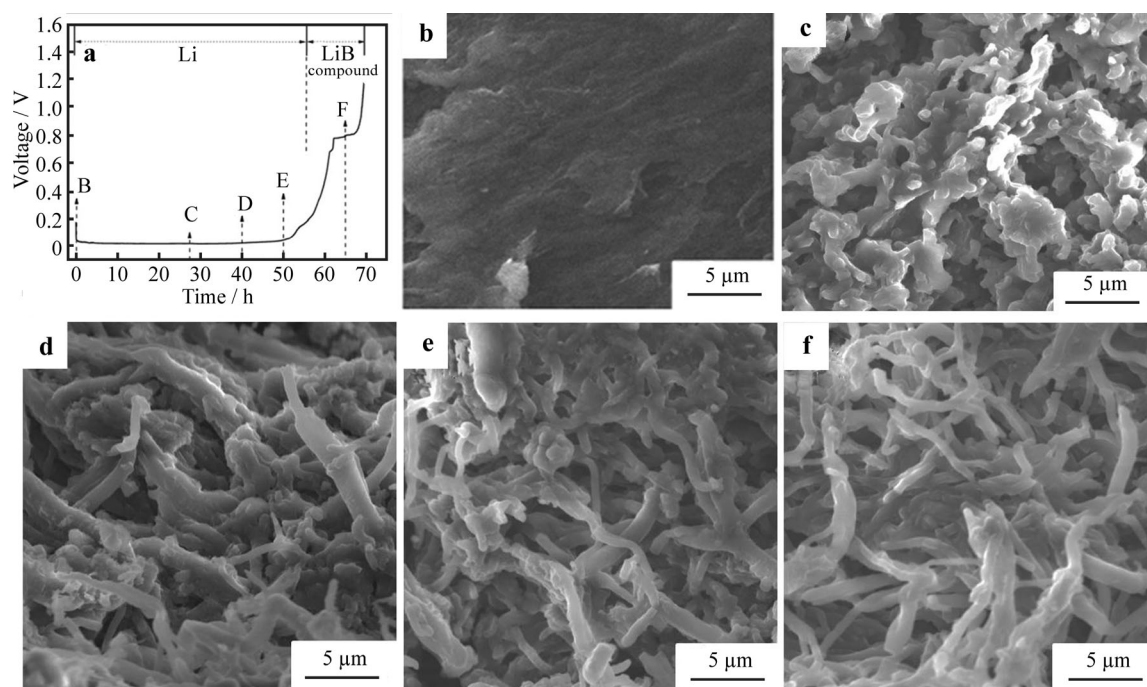


Fig. 3 a Electrochemical curve of Li-B alloy cycled at current density of $0.5 \text{ mA}\cdot\text{cm}^{-2}$; SEM images of Li-B alloy anode b before electrochemical cycling and after Li dissolution for c 27 h, d 40 h, e 50 h and f 65 h

the deposited Li formed the LiB compound again and the other part was deposited as metal Li in the pores of the fibrous skeleton. The LiB compound has three discharge platforms of 0.46, 0.69 and 0.80 V [23]. If the Li-B alloy is discharged to a voltage higher than 0.80 V, the structure of the LiB compound will change and the Li dissolution/deposition will be irreversible.

The morphological evolution of the Li-B alloy anode during Li dissolution is shown in Fig. 3b-e. At the beginning of Li dissolution (Fig. 3c, d), the LiB compound fibers were surrounded by pure Li. With the further electrochemical dissolution of Li, the interlaced and smooth LiB compound skeleton was exposed. The diameter of the LiB fibers is $\sim 0.5\text{--}1.0 \mu\text{m}$ (Fig. 3e, f). The LiB compound participates in the discharge because the over-dissolved Li has some influence on the morphology of LiB fibers. It can be seen that some LiB fibers became thinner and the ends of the fibers became sharp.

Previous reports demonstrated that the Li-deficient LiB compound could attain a Li-rich state when immersed in the Li liquid [30]; the Li-B alloy could still maintain the apparent solid state at high temperatures, and liquid Li did not flow out [25], which indicated a good lithiophilic property of the LiB compound. Therefore, Li was preferentially deposited on the surface of the LiB compound. Figure 4 illustrates the morphology of the Li-B alloy after Li deposition. Compared with Fig. 3, the exposed LiB fibers after delithiation have been completely covered by Li

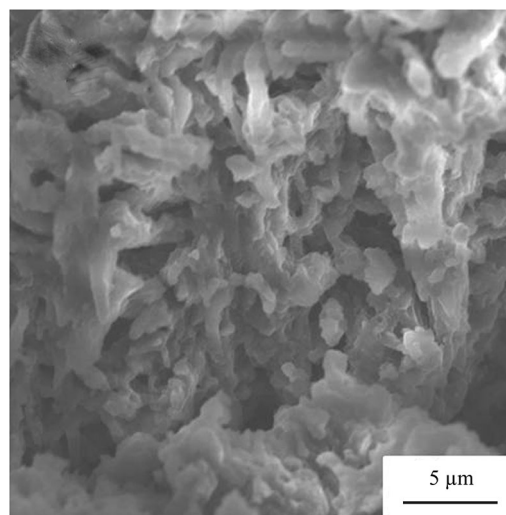


Fig. 4 SEM image of Li-B alloy anode after Li deposition

during Li deposition. It can be seen that Li would preferentially deposit on the surface of LiB fibers due to its good lithiophilic property and conductivity [27], which can suppress the generation of dendrites on the anode surface.

3.3 Electrochemical performance of Li-B alloy

To study the electrochemical performance of the Li-B alloy anode, the asymmetric cells with Li-B alloys of different B contents were assembled as working electrodes

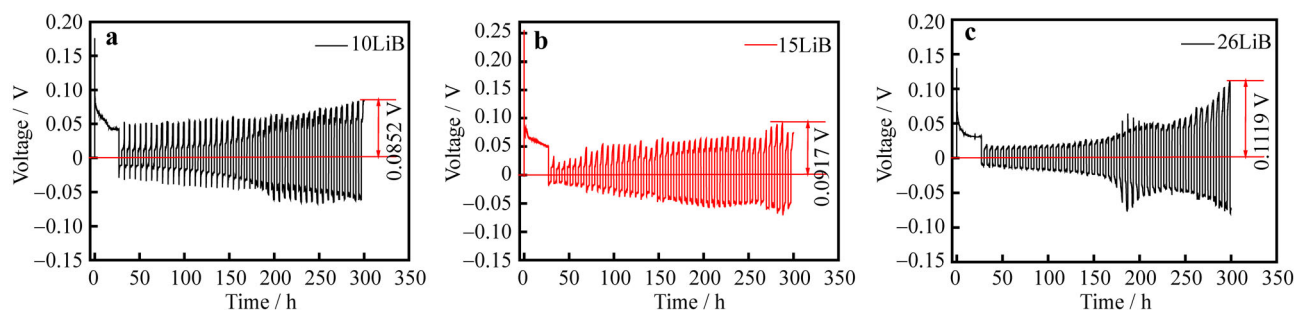


Fig. 5 Galvanostatic charge–discharge curves of Li–B/Li asymmetric cells for **a** 10LiB, **b** 15LiB and **c** 26LiB

and cycled by galvanostatic charging/discharging (Fig. 5). Before cycling, the Li–B alloy was delithiated for 27 mAh in the first cycle. The Li–B alloy was then cycled at a current density of $0.5 \text{ mA}\cdot\text{cm}^{-2}$ with a capacity of $3 \text{ mAh}\cdot\text{cm}^{-2}$ in each cycle. The LiB skeleton with high B content had a larger specific surface area, and the actual current density was reduced correspondingly; therefore, 26LiB had a low overpotential in the initial cycles. After cycling for 300 h, the overpotentials of the Li–B alloys were 0.0852, 0.0917 and 0.1119 V for 10LiB, 15LiB and 26LiB, respectively.

To study the interfacial changes, the EIS of the Li–B alloys after different cycles was performed, as shown in Fig. 6. The EIS results are fitted by an equivalent circuit as shown in the insets of Fig. 6a, d. R_1 is the bulk resistance of the cell, which reflects a combined resistance of the electrolyte, separator and electrodes. R_2 and R_3 are the SEI film resistance and charge–transfer resistance, respectively [6]. The skeleton became loose as the B content decreased. This is because the amount of the LiB compound decreased, and the rate of increase in the impedance of the SEI film gradually decreased. It is observed from Fig. 6a–c that the internal impedance of the cell assembled with

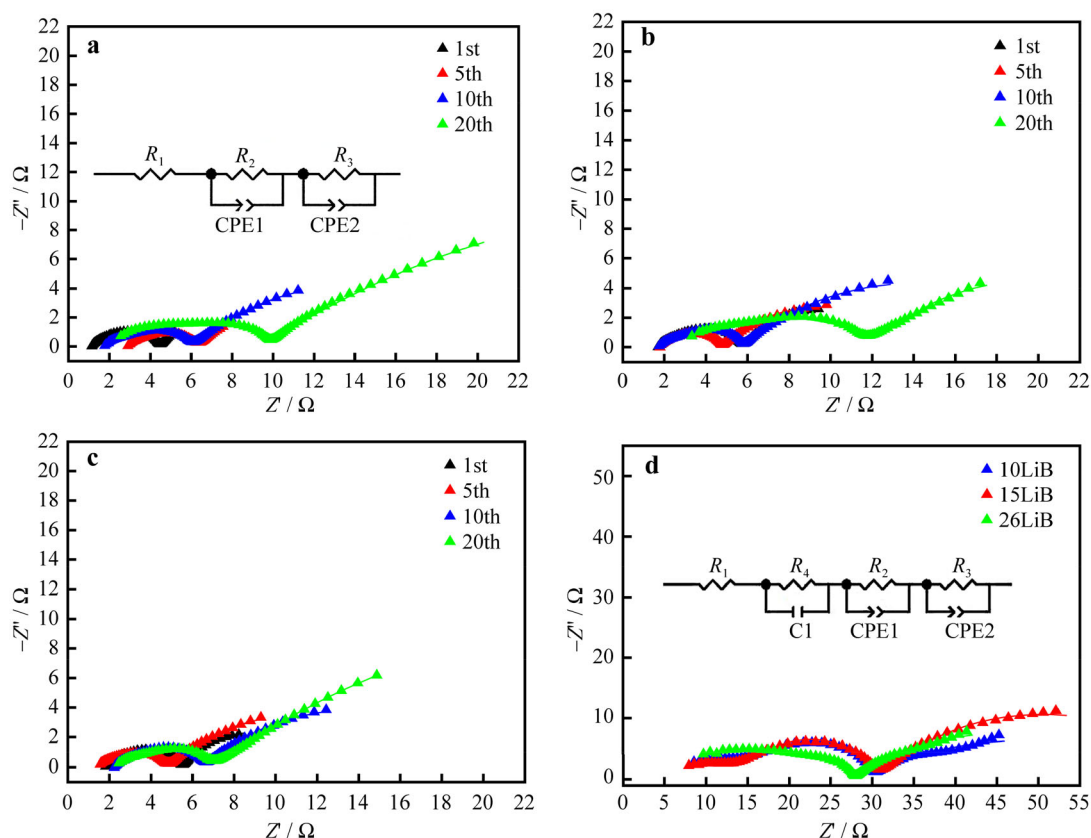


Fig. 6 Electrochemical impedance spectra of **a** 26LiB, **b** 15LiB, and **c** 10LiB after different cycles, and **d** Li–B alloys with different B contents after 50 cycles (Z' : real part of impedance; Z'' : imaginary part of impedance)

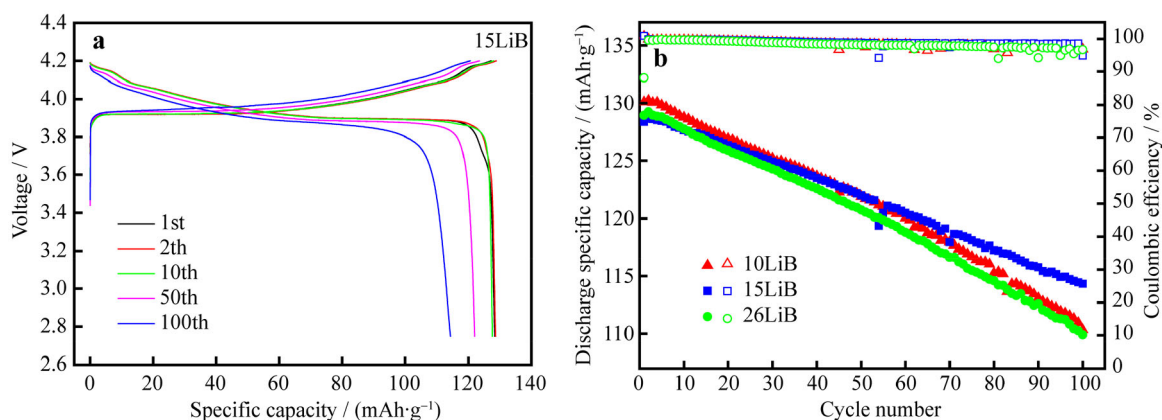


Fig. 7 **a** Charge–discharge curves of 15LiB and **b** cycle performance of coin-type full cell

10LiB displays the smallest changes after cycling. In addition, the EIS spectrum of the cells after 50 cycles was slightly different from the previous EIS spectrum, which could be fitted by another equivalent circuit as shown in the inset of Fig. 6d with a pure capacitor (C_1) and corresponding resistance (R_4) added to represent a smoother interface. It is believed that the semicircle from high to intermediate frequencies shown in Fig. 6d is because of the accumulation of the dead SEI depositions, causing Li to deposit on the surface of the anode rather than between the skeletons. The decrease in the boron content will reduce the amount of LiB compound fibers, which increases the distance between the LiB fibers frameworks and eventually alleviates the thickening of the SEI film during Li dissolution/deposition. Therefore, the interface resistance is the lowest for the Li–B alloy with 10 wt% B content. The results also indicate that decreasing the amount of B is an effective way to alleviate the generation of the dead SEI depositions to some extent.

The full cell was tested in the voltage range of 2.75–4.2 V, at a current density of 0.2C ($1C = 148 \text{ mAh}\cdot\text{g}^{-1}$). Figure 7a illustrates the charge–discharge curves of 15LiB, and Fig. 7b displays the discharge specific capacity and coulombic efficiency curves of the full cell with three Li–B alloys as the anodes and $\text{Li}_{(1-x)}\text{CoO}_2$ as the cathode. The voltage platform was between 3.8 and 3.96 V. It can be seen that the cycle performance of the Li–B alloy was relatively stable, and the capacity retention rate was between 85% and 89% after 100 cycles. Among them, the discharge specific capacity of the $\text{Li}_{(1-x)}\text{CoO}_2$ ||15LiB alloy full cell can reach $\sim 115 \text{ mAh}\cdot\text{g}^{-1}$ after 100 cycles, and the coulombic efficiency was above 97% during the whole cycling. This indicates that the reduction in the B content could improve the cycling performance of the Li–B alloy to some extent. However, the performance of 10LiB is not as good as that of 15LiB. This may be attributed to the reduction in the B content, which leads to a loose

compound skeleton, resulting in a large volume change in the negative electrode during the cycling. When the Li–B alloy undergoes a dramatic volume reduction, the internal porosity of the alloy will decrease greatly and some Li may not deposit into the pores between the skeleton in the subsequent cycling process. Finally, the Li will accumulate on the surface of the negative electrode and cause a decrease in the cycle performance.

4 Conclusion

The electrochemical characteristics of the Li–B alloys with three different B contents were explored. It was demonstrated that the overpotentials of the asymmetric cells decreased with the reduction in the B content. The interfacial impedance of the Li–B alloy with 10 wt% B is the lowest among the three Li–B alloys. However, the Li–B alloy with appropriate B content (15LiB) demonstrated the best electrochemical performance because the LiB skeleton that is very loose and with less content of B may not effectively suppress the large volume change during cycling. This suggests that the ideal performance of the Li–B alloy anode can be obtained via tuning the B content for next-generation high energy density lithium batteries.

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